

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1991

Cabin Creek 345 kV
September 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1991. The Interconnection Customer proposes a 87.5 MW solar plus storage facility interconnecting at the Cabin Creek 345 kV substation in the West Virginia area. The request was received on January 25, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 548 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,420,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lerdo to Tidd 500 kV	Filer Substation to Tidd 500 kV	103.1	2,594	6.0
Mattoon E to Lexington Health Ctr 138 kV	Coulee Ave to Amos 345 kV line	102.5	216	5.0
Dixieland Prison to Lexington Health Ctr 138 kV	Coulee Ave to Amos 345 kV	102.1	221	5.0
New Gulf to Dixieland Prison 138 kV Line	Amos to Coulee Ave 345 kV Line	101.5	225	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lerdo to Tidd 500 kV	Reconductor line	\$39,408,000	1% (\$324,000)
Mattoon E to Lexington Health Ctr 138 kV	Reconductor line	\$3,670,000	25% (\$928,000)
Dixieland Prison to Lexington Health Ctr 138 kV	Reconductor line	\$2,928,000	25% (\$740,000)
New Gulf to Dixieland Prison 138 kV Line	Reconductor line	\$1,693,000	25% (\$428,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$2,420,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.